

The **Rural Student Management System** is a web-based application developed using **Flask** and **Python**. It provides a centralized platform for managing academic activities such as attendance, study materials, homework, messaging, and notices.

The system is designed to improve communication between teachers and students while reducing manual work and increasing efficiency.

Features

Teacher Features

- Teacher registration and login
- Manage student records
- Mark attendance
- Upload study materials
- View and grade homework
- Reply to student messages
- Post notices

Student Features

- Student registration and login
 - View attendance
 - Download study materials
 - Submit homework
 - Check grades
 - Send messages to teachers
 - View notices
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Technologies Used

- **Backend:** Python, Flask
 - **Frontend:** HTML, CSS
 - **Data Handling:** Pandas
 - **Database:** CSV files
 - **File Handling:** Werkzeug
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Project Structure

student-teacher-management/

```
|
├── app.py
├── requirements.txt
├── README.md
|
├── data/
├── uploads/
├── templates/
├── static/
└── utils/
```

How It Works

1. Users register as teacher or student
 2. Data is stored in CSV files
 3. Users log in using credentials
 4. Dashboard displays relevant information
 5. Users perform actions (attendance, upload, messaging, etc.)
 6. Data updates in real-time
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Security Features

- Session-based authentication
 - Role-based access control
 - Secure file uploads
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Advantages

- Simple and lightweight system
 - Easy to use interface
 - No need for complex database
 - Efficient data management
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Future Enhancements

- Integration with MySQL or MongoDB
- Mobile application support

- Email/SMS notifications
 - Advanced authentication system
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References

- [Flask Documentation](#)
- [Pandas Documentation](#)
- [Python Official Documentation](#)